ECONOMETRICS OF COMMODITY AND ASSET PRICING

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The main objective of this course is to propose discrete time methods for pricing commodities and financial assets. These methods are based on four pillars:

i) Financial pillar - the absence of arbitrage opportunity (AAO) principle.

ii) Mathematical pillar - the (multi-horizon) Laplace transform.

iii) Probabilistic pillar - the class of discrete-time affine processes.

iv) Statistical pillar - the non-linear state-space models.

The methods are first applied to commodity markets in order to price forward and futures contracts, taking into account the convenience yields. A modeling of spot and forward electricity prices is developed, as well as a simultaneous modeling of several commodity markets. Then the pricing methods are applied to various financial domains: sovereign and corporate bonds with possible switching regimes and/or zero lower bound spells, interest rate derivatives , default and illiquidity risks, quadratic and Wishart interest rate models , credit event pricing , option pricing including conditional heteroskedasticity or stochastic volatilities and/or switching regimes. The statistical problems of inference, filtering, smoothing and prediction are treated.

Chapter I - DISCRETE TIME AFFINE PROCESSES

- Information in the Economy: the factors.
- Dynamic Models.
- Some properties of the Laplace transform.
- Affine (or Car) Processes.
- Extended affine processes.
- Truncated Laplace transforms of affine processes.

Chapter II - PRICING AND RISK NEUTRAL DYNAMICS

- Stochastic Discount Factor: equilibrium approach.
- Stochastic Discount Factor: absence of arbitrage opportunity approach.
- Exponential-Affine SDF.
- The Risk Neutral Dynamics.
- Typology of Econometric Asset Pricing Models.

Chapter III - FORWARDS, FUTURES, DIVIDENDS, COMMODITY PRICING AND CONVENIENCE YIELDS

- Forward contracts, forward prices.

- Futures contracts, futures prices.
- Convenience yields.
- Pricing with affine models.
- A Gaussian VAR model.
- Applications.

Chapter IV - MODELING OF SPOT AND FORWARD ELECTRICITY PRICES

- Specific features of electricity markets (Spikes, Seasonality, Non-Storability, Continuous Delivery).
- Modelling approach.
- Pricing.
- Application.

Chapter V - SIMULTANEOUS MODELING OF SEVERAL COMMODITY MARKETS

- A general affine framework.
- A Gaussian VAR model.
- A regime switching Gaussian VAR Model.

Chapter VI – BASIC CONCEPTS IN FIXED INCOME ANALYSIS

- Discount Factors.
- The Term Structure of Interest Rates.
- From Zero-Coupon Bonds to Coupon Bonds.
- From Coupon Bonds to Zero-Coupon Bonds.
- Yield to Maturity and Return.
- Extracting Yield Curves from Bond Prices.

Chapter VII - GAUSSIAN AFFINE TERM STRUCTURE MODELS

- Introduction.
- Affine Term Structure Models (ATSMs).
- Univariate Gaussian ATSMs.
- Bivariate Gaussian ATSMs.
- Multivariate Gaussian ATSMs.
- Back modelling of Gaussian ATSMs.
- Empirical analysis of Gaussian ATSMs.

- An alternative estimation procedure.
- Principal Component Analysis of the Yield Curve.

Chapter VIII - A NEW PERSPECTIVE ON GAUSSIAN AFFINE TERM STRUCTURE MODELS – Part I

- Introduction.
- A Gaussian Model.
- Information in the cross-section.
- The role of invertibility in empirical analysis.
- A two-factor example.
- A Canonical hidden-factor model.

Chapter IX - A NEW PERSPECTIVE ON GAUSSIAN AFFINE TERM STRUCTURE MODELS – Part II

- Introduction.
- A Canonical Gaussian ATSMs.
- Observational Equivalence.
- About Unspanned Macro Variables.

Chapter X - SWITCHING REGIMES GAUSSIAN AFFINE TERM STRUCTURE MODELS

- Switching Regimes Car processes.
- Univariate Switching Regimes Gaussian ATSMs.
- Multivariate Switching Regimes Gaussian ATSMs.
- Empirical analysis.

Chapter XI - NON-NEGATIVE AFFINE TERM STRUCTURE MODELS

- Autoregressive Gamma ATSMs.
- Switching Regimes Autoregressive Gamma ATSMs.
- Wishart and Quadratic TSMs.
- Non-negative ATSMs.

Chapter XII - ESTIMATION OF AFFINE ASSET-PRICING MODELS

- State-Space models.
- Kalman-filter-based approaches.

- Inversion technique.
- Dealing with unobserved regimes.
- The persistence problem.

Chapter XIII - PRICING CREDIT AND LIQUIDITY RISKS

- Notations.
- Pricing under standard assumptions.
- Pricing illiquid bonds.
- Relaxing the classical assumptions.
- General affine credit-risk framework.
- CDS pricing in the general framework.
- Top-Down approach.

Chapter XIV - ECONOMETRICS OF OPTION PRICING

- Security Market Models.
- Truncated Laplace Transforms (TLT) and Transform Analysis.
- Direct Modelling of Conditionally Normal Processes.
- Back Modelling of Conditionally Normal Processes.
- Direct Modelling of Mixed-Normal Processes.
- Direct Modelling of Conditionally Spline Dynamics.
- Back Modeling of Regime Switching Dynamics.
- Back Modeling of Stochastic Volatility Dynamics.
- Back Modeling of Regime Switching GARCH Dynamics.
- Back Modeling of Regime Switching IG GARCH Dynamics.
- A Component Volatility GARCH Model.

Chapter XV – ASSET PRICING WITH SECOND-ORDER ESSCHER TRANSFORM

- Introduction.
- The Second-Order Esscher Transform.
- Conditionally Gaussian Economies.
- Calibrating the Second-Order GARCH Option Pricing Model.

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